CrystalRL — Crystal Clear Reinforcement Learning

Publication proposal: what does an agent gain when interpretability is built in rather than explained after the fact?

Ivan Pavliuk

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Abstract

The field that studies how to explain reinforcement-learning (RL) agents has, by its own account, a measurement problem: interpretability is routinely claimed and rarely measured, and the surveys say so explicitly. We completed a study designed so the claim *could lose*. One trading agent was built three times: first as a strong conventional agent whose decisions come from an unreadable learned state (a "black box"); then the black box was kept and the best available *after-the-fact* explanation program was run over it; finally the agent was rebuilt so that its entire memory is one *named* number — the probability that the market is in a falling regime — and its policy is a rule readable as a sentence. All were scored with one number — generations 1 and 2 share a score, because the after-the-fact reading is the black box's only route to a human-sized story — CrystalScore, which multiplies three measurable properties: whether setting the agent's named internal belief to a chosen value changes its behavior exactly as the name promises (faithfulness), whether a nine-rule story can reproduce its behavior without losing performance (simulatability), and whether the mechanism survives retraining (stability). The headline findings, on real market data: commands "work" on the black box too, so command success does not distinguish designs; the axis that does distinguish them is simulatability ($0.244 \rightarrow 0.92$); and the rebuilt agent's one-sentence rule passed its single pre-registered test on 629 days of unseen data, beating buy-and-hold on risk-adjusted return with a smaller worst loss. We ask approval to write the full paper.

1 The problem, as the field itself states it

(1) Interpretability in RL is asserted, not measured. Successive surveys in ACM Computing Surveys find that evaluations of "explainable RL" are ad hoc and not comparable across papers, with no agreed metrics and few human studies [Milani et al., 2024, Vouros, 2023]. A 2025 evaluation paper states it plainly: there is "no clear definition of policy interpretability, i.e., no clear metrics", and claims are routinely made without measurement [Kohler et al., 2025].

(2) The imported metrics are themselves suspect. The measures RL borrows from natural language processing have been shown, in their home field, to measure a model's self-consistency rather than whether an explanation reflects the real computation [Parcalabescu and Frank, 2024]; and the most popular visual technique for RL agents — saliency maps, heatmaps of where the agent "looks" — was shown by counterfactual analysis to be "exploratory, not explanatory" [Atrey et al., 2020]. Any new metric must therefore be *causal*: test explanations by intervening on the agent, not by admiring them.

(3) A concrete missing artifact: a writable belief. For agents that act under uncertainty, theory says their recurrent memory approximates a *belief state* — an internal probability estimate over what is currently true in the world [Lambrechts et al., 2022, Kaelbling et al., 1998]. The literature can *detect* such beliefs inside trained networks; what does not exist is a deployed agent whose belief is a named quantity a human can read, *write* (set to a chosen value and have behavior follow, exactly as the name promises), and certify. That interface is precisely what oversight of a self-modifying agent requires — the subject of our companion paper (*Hello Crystal*, prepared concurrently).

2 The research question

What does an RL agent measurably gain — on a metric that can lose — when interpretability is built in, rather than explained after the fact?

Answering it requires all three objects on one task, scored identically: a competent black box, its best after-the-fact reading, and a successor rebuilt to be readable. We built all three.

3 The three agents, in plain words

The task is daily portfolio management: each day the agent decides how much capital is in stocks and how much in cash, and which stocks. All three generations trade the same U.S. blue-chip daily stock panel (Dow Jones constituents plus a cash account, data through 2026); the final pre-registered test window is January 2024 through July 2026 (629 trading days).

Generation 1 — a strong conventional agent (CHRL, "Constrained Hierarchical RL"). A deep-RL agent whose decisions are factored into three readable stages (overall risk level; sector groups; individual stocks), each with hard safety rails (e.g., exposure can only change gradually; new buying is blocked until recovery evidence appears). The stages are readable; the *core* is not: between input and action sits a learned 64-dimensional internal state with no names on its axes, tuned by ∼214 configuration parameters (thresholds, penalty weights, limits) — none of which names a belief or an intention. In standard walk-forward validation (train on a period, test on the next, roll forward) it earns 93% of the buy-and-hold benchmark's Sharpe ratio (annualized return divided by its volatility) with about half its maximum peak-to-trough loss. An ablation — rebuilding the agent with each ingredient removed — shows honestly where that comes from (Figure 1): the safety rails, not the staged structure, carry the risk result; structure alone actually hurts.

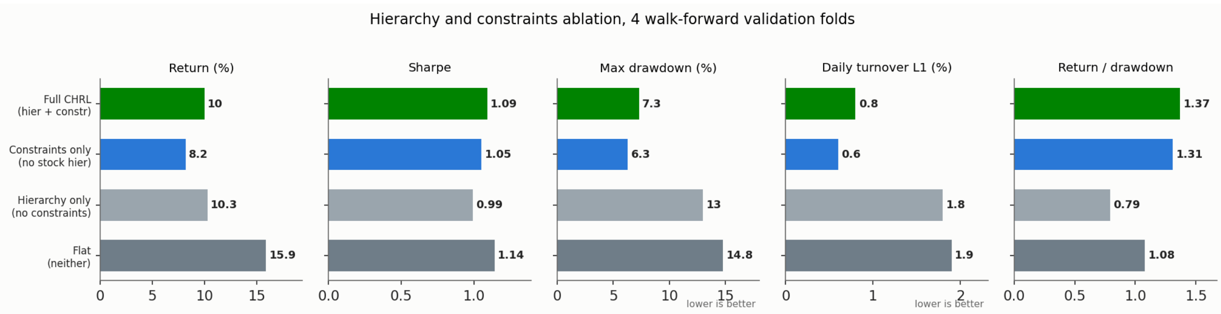

Figure 1: Rebuilding generation 1 four ways, each tested on 4 walk-forward folds. Bars show, left to right per panel: the full agent; rails without staged structure; structure without rails; neither. Panels, left to right: annual return (%); Sharpe ratio; maximum drawdown (%); daily trading volume (% of the book turned over); and return/drawdown — annual return divided by the worst peak-to-trough loss, the risk-adjusted summary. The rails halve the worst loss and the trading volume; structure alone is the worst variant; only the combination gets the best ratio.

Generation 2 — the best after-the-fact reading of that black box. With the trained agent frozen, we logged its internal states, clustered them into eight recurring "behavior patterns", labeled each from the trading record, and then *intervened*: nudged the internal state toward or away from a chosen pattern and passed the edited state through the unchanged action network, so any behavioral change is attributable to the nudge. The central methodological lesson: a nudge toward a *random* target also "improves" results (+0.55 percentage points of cumulative return over the frozen 2022–23 test window) — gentle noise shakes a slightly suboptimal policy off its habits — so every claimed effect is reported *minus* its random-nudge control. Under that discipline, targeted edits genuinely work (+0.19 to +0.30 percentage points, control-adjusted).

And the same program measures where after-the-fact reading stops: a story limited to nine rules reproduces only 24.4% of the deployed agent's decisions; there is no way to bound what else an internal edit touches; and the ∼214 knobs contain no command a human could name (Figure 2).

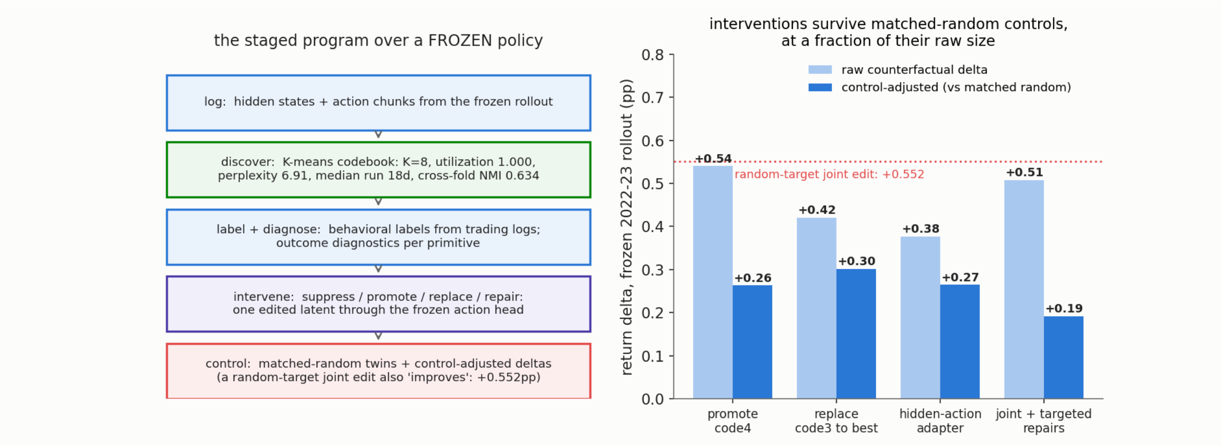

Figure 2: The after-the-fact program. Left: its pipeline over the frozen agent — log internal states, discover eight recurring behavior patterns, label them, intervene, and control every claim against random-target nudges. Right: the flagship effects, raw and after subtracting the random-nudge control (the dotted line is the random nudge itself — the reason controls are mandatory).

Generation 3 — the agent rebuilt readable (CRYSTAL-1). Each measured wall becomes an architectural decision. The agent's *entire* memory is one named number: the probability b_t that the market is in a falling ("bear") regime, updated daily by an explicit two-regime Bayesian filter,

$$b_t = \frac{f_{\text{bear}}(r_t)\,\tilde{b}_t}{f_{\text{bear}}(r_t)\,\tilde{b}_t + f_{\text{bull}}(r_t)\,(1-\tilde{b}_t)}, \qquad \tilde{b}_t = \pi_{BB}\,b_{t-1} + \pi_{GB}\,(1-b_{t-1}), \tag{1}$$

where r_t is the day's market return, f_{bear} and f_{bull} are the statistical profiles of daily returns in falling and rising regimes, $\tilde{b}_t$ carries yesterday's belief forward through the regimes' persistence probabilities (π_{BB}: bear stays bear; π_{GB}: bull turns bear — B for bear, G for the good regime), and the ratio is Bayes' rule. Because b_t is the *only* memory, the effect of writing to it is bounded by construction — the guarantee generation 2 could test for but never provide. The policy is a small decision tree over $[b_t,\ \text{holdings},\ \text{time}]$ at full performance parity (the readable form gives up none of the unconstrained version's performance); in its certified form — "certified" meaning it passed the frozen statistical exam of the companion paper, including placebo controls, before its one pre-registered test — it degenerates to one sentence: *if* $b_t > 0.66$, *cut stock exposure to* 74%.

4 The measurement: a score that can lose

$$\textsc{CrystalScore} = F \times S \times St, \tag{2}$$

computed identically for every agent, where each factor is between 0 and 1:

- F (**faithfulness**): the fraction of *command tests* that succeed — set the belief to a chosen value and check the action changes exactly as the name promises, judged against what an ideal holder of that belief would do (the optimal action given that belief under the world model that defines the regimes). This is a causal test, per the standard of §1(2), not a visual inspection.
- S (**simulatability**): the fraction of the agent's decisions reproduced by a story of at most nine named rules, under the constraint that the story-following agent *performs as well as the original* — a story that only explains a crippled version of the agent is cheating. Nine rules

is a working-memory-sized budget; the measure is a program-executable proxy for human simulatability.

- *St* (**stability**): the fraction of independent retrainings (different random seeds) in which the same mechanism reappears.

One calibration lesson is itself a result. In a synthetic environment where the optimal rule is known, we trained an agent to follow that rule (97% action agreement) — an agent that *is* the rule for all practical purposes. Our first faithfulness test still returned $F = 0$ on it, because the test's fixed-size belief nudges never crossed the rule's decision threshold. The repaired test writes contrasting values on both sides of the threshold; under it, taught agents in the calibration environment read $F = 0.77$–0.83 and untaught ones still read 0, while the deployed agents of Table 1 read $F = 1.0$. *Interpretability metrics are instruments, and instruments need calibration.*

5 Results

result	what was measured	evidence / protocol
one score, three generations	black box + its reading: F=1.00 × S=0.244 × St=0.619 = **0.151**; rebuilt readable: F=1.00 × S=0.92 × St=1.00 = **0.92** (for the black box, stability = the share of retrainings in which the same discovered mechanism recurs)	identical protocol, real market data, nine-rule story budget
the discriminating axis	command tests succeed on *both* ($F = 1.0$) — command success cannot tell the designs apart; the story axis can: 24.4% vs 92% of decisions reproduced at equal performance	the paper's central finding
a writable belief	writing the belief moves behavior exactly as named in 100% of contrast tests; the state named "bear" is verifiably the state where derisking is optimal; the mechanism reappears in 10/10 retrainings; a proposed belief boost failed the pre-set non-inferiority threshold — no edit may degrade the certified objective — and was correctly *refused*	causal command tests, not visual inspection
the certified sentence	"if $b_t > 0.66$, cut stock exposure to 74%" beat buy-and-hold on the pre-registered primary endpoint — risk-adjusted improvement vs a matched twin, observed $z = 2.06$ (∼95% confidence) — with Sharpe 1.46 vs 1.39 and a smaller worst loss (−12.9% vs −15.5%)	one pre-registered test on 629 unseen trading days
after-the-fact reading: real but bounded	targeted internal edits help by +0.19–0.30 percentage points *after* subtracting the random-nudge control (+0.55 pp raw — controls are mandatory); the nine-rule story ceiling is 24.4%	frozen agent, matched-random controls
where the risk result comes from	return/drawdown: full agent 1.37; rails only 1.31; structure only 0.79; neither 1.08	4-way rebuild, 4 walk-forward folds
metric calibration	the strict command test read $F = 0$ on an agent that *is* the rule (97% clone); the repaired test reads 0.77–0.83 on taught agents and still 0 on untaught ones	a designed environment with known ground truth

Table 1: Results at a glance. Sharpe ratio = annualized return divided by its volatility (risk-adjusted return); worst loss = maximum peak-to-trough decline; pp = percentage points; walk-forward = train on one period, test on the next, roll forward.

Beyond the table, one comparison summarizes the paper (Figure 3): the two architectures scored on the same three axes. Command success (F) is 1.0 for both — a black box can be steered too, which is why steering demonstrations, the most common evidence in this literature, do not establish interpretability. What separates the designs is whether a human-sized story can *be* the agent: 24.4% versus 92% of decisions reproduced at equal performance. Building the concepts in — rather than discovering them afterwards — is worth a factor of ∼4 on the story axis and a factor of 6 on the combined score, at zero cost in trading performance.

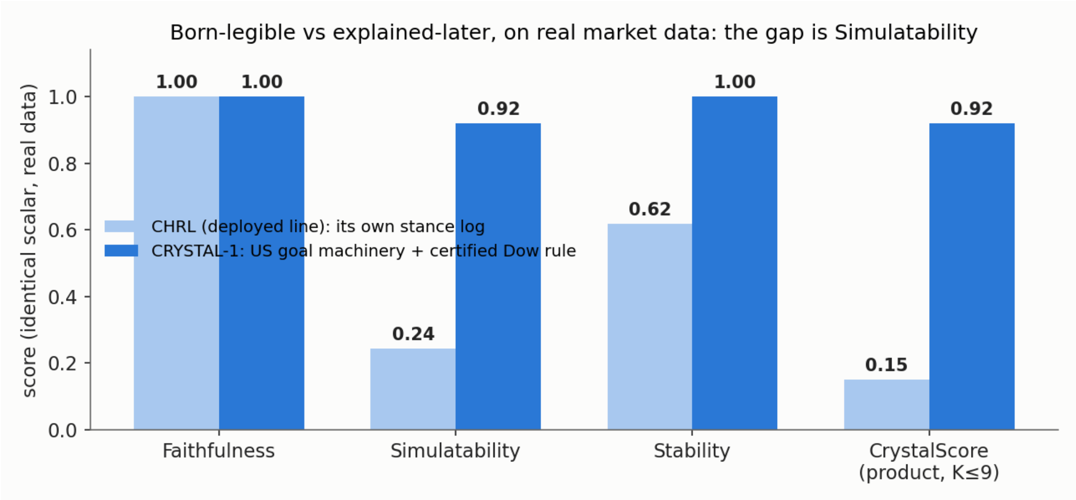

Figure 3: The same score on both architectures, real data only. Left group of bars: faithfulness (command tests), identical for both. Middle: simulatability (the nine-rule story), the discriminating axis. Right: stability across retrainings, and the combined CrystalScore.

6 The claim, and the request

The paper will claim: **(i)** the first quantitative comparison of a black box, its best after-the-fact explanation program, and a rebuilt-readable successor under one metric that can lose, on a real high-stakes task — the evaluation instrument the surveys of §1 explicitly request; **(ii)** the finding that command success does not discriminate architectures while parity-constrained simulatability does; **(iii)** the first deployed agent whose belief state is a named, writable, certified interface — including the safety bar that refused an unsafe write; **(iv)** the calibration lesson that interpretability metrics are instruments requiring ground-truth surfaces. A companion paper (*Hello Crystal*) covers the statistically governed self-improvement loop that modifies this agent; its safety precondition — *an agent may change itself only to the degree it can be read* — is exactly what this paper makes measurable. All results are complete, logged, and audited; this document requests approval to write the full paper.

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